

Background Process For loops

```
for (let i = 1, i <= 3, i++) {
  log("Hello");
}
```

init	condition	logic	update
$i = 1$	$i <= 3$	log("Hello")	$i++$
$i = 1$	$1 <= 3 (T)$	Hello	$i+1 = 1+1 = 2$
$i = 2$	$2 <= 3 (T)$	Hello	3
$i = 3$	$3 <= 3 (T)$	Hello	4
$i = 4$	$4 <= 3 (F)$	X	X

→ even number in range (1, 10)

```
for (let i = 1, i <= 10, i++) {
  if (i % 2 == 0) {
    console.log(i);
  }
}
```

ini	cond	logic	update
$i = 1$	$i <= 10$	$i \% 2 == 0$	$i = i + 1$
$i = 1$	$1 <= 10 (T)$	$1 \% 2 == 0 (F)$	$i = 1 + 1 = 2$
$i = 2$	$2 <= 10 (T)$	$2 \% 2 == 0 (T)$	3
$i = 3$	$3 <= 10 (T)$	$3 \% 2 == 0 (F)$	4
$i = 4$	$4 <= 10 (T)$	$4 \% 2 == 0 (T)$	5
$i = 5$	$5 <= 10 (T)$	$5 \% 2 == 0 (F)$	6
$i = 6$	$6 <= 10 (T)$	$6 \% 2 == 0 (T)$	7
$i = 7$	$7 <= 10 (T)$	$7 \% 2 == 0 (F)$	8
$i = 8$	$8 <= 10 (T)$	$8 \% 2 == 0 (T)$	9
$i = 9$	$9 <= 10 (T)$	$9 \% 2 == 0 (F)$	10
$i = 10$	$10 <= 10 (T)$	$10 \% 2 == 0 (T)$	11
$i = 11$	$11 <= 10 (F)$	X	X

#Print odd number from ~~1~~ 1 to 5

```
for(i=1; i ≤ 5; i++) {
```

```
    if(i % 2 != 0) {
```

```
        Print(i)
```

```
    }
```

Init	Con	logic	Updation
i = 1	i ≤ 5	i % 2 != 0	i = i + 1
1	1 ≤ 5 (T)	1 % 2 != 0 (T)	2
2	2 ≤ 5 (T)	2 % 2 != 0 (F)	3
3	3 ≤ 5 (T)	3 % 2 != 0 (T)	4
4	4 ≤ 5 (T)	4 % 2 != 0 (F)	5
5	5 ≤ 5 (T)	5 % 2 != 0 (T)	6
6	6 ≤ 5 (F)	X	X

#Print the reverse number 5 to 1

// 5 4 3 2 1

```
for (i = 5, i ≥ 1, i--) {
```

```
    console.log(i)
```

```
}
```

init	cond	log	update
$i = 5$	$5 >= 1 (T)$	5	$i = 5 - 1 = 4$
4	$4 >= 1 (T)$	4	3
3	$3 >= 1 (T)$	3	2
2	$2 >= 1 (T)$	2	1
1	$1 >= 1 (T)$	1	0
0	$0 >= 1 (F)$	X	X

Factorial of numbers using for loop

let $n = 3$

let $mul = 1$

for (let $i = 1, i <= n, i++$) {
 $mul = mul * i$
 $console.log(mul)$

3

Init	cond	log	update
$i = 1$	$i <= 3$	$mul = mul * i$	$i = i + 1$
$i = 1$	$1 <= 3 (T)$	$1 \times 1 = 1$	2
$i = 2$	$2 <= 3 (T)$	$1 \times 2 = 2$	3
$i = 3$	$3 <= 3 (T)$	$2 \times 3 = 6$	4
$i = 4$	$4 <= 3 (F)$	X	X